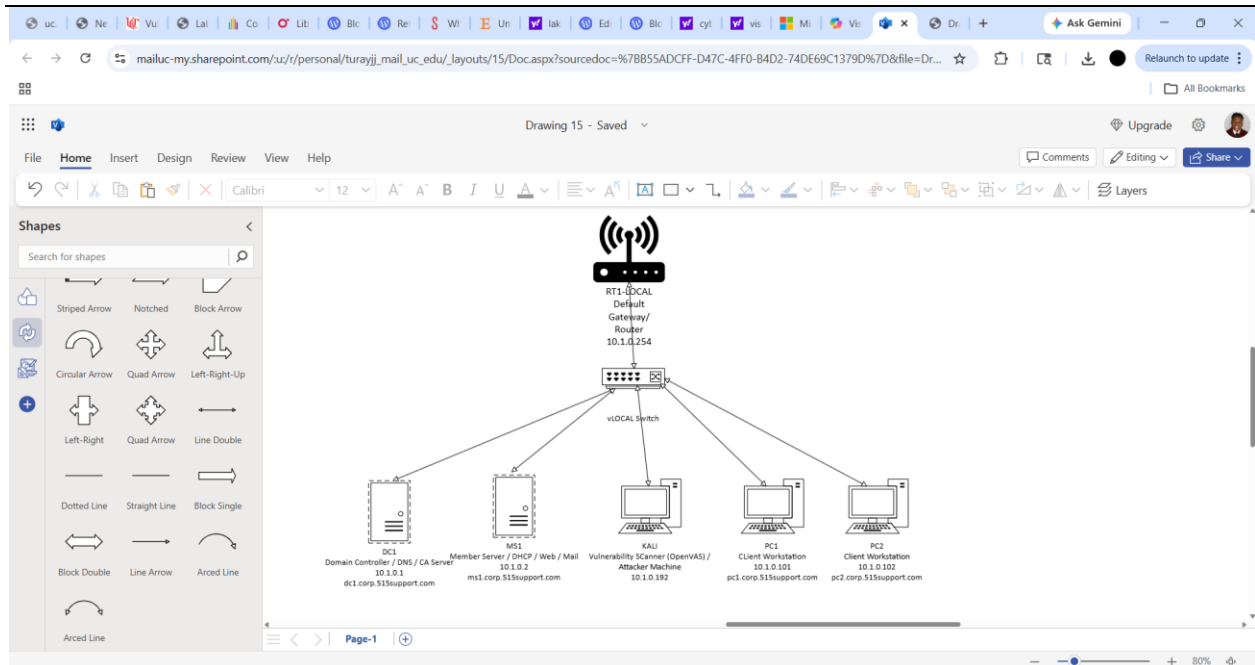


Network Security Lab Report

Lab Title: Vulnerability Scanning Lab

Network Diagram(s)



System Information

VM Name	Operating System	IP Address	FQDN	Role/Purpose
RT1-LOCAL	Vyos/Linux Router	10.1.0.254	N/A	Default gateway/router for the vLOCAL network
DC1	Windows Server 2016	10.1.0.1	Dc1.corp.515support.com	Domain Controller, DNS Server, and Certificate Authority
MS1	Windows Server 2016	10.1.0.2	Ms1.corp.515support.com	Member server, DHCP Server, IIS web server, Mail server

KALI	Kali Linux	DHCP assigned 10.1.0.192	N/A	Vulnerability Scanning/ attacker machine
PC1	Windows 10	DHCP assigned 10.1.0.101	Pc1.corp.515support.com	Client Workstation
PC2	Windows 7	DHCP assigned 10.1.0.102	Pc2.corp.515support.com	Client Workstation

Objective

- The objective of this lab is to understand how vulnerability scanning actually works in a real environment by using tools on a Kali Linux system. Instead of just learning commands, the goal is to see how a security analyst or attacker can identify weaknesses across a network, analyze the results, and understand how those vulnerabilities could be exploited. This lab also helps build a better understanding of how systems like servers and workstations can be assessed and secured in a real-world network.

Discussion

- In this lab, I used Kali Linux and OpenVAS to scan the network and analyze vulnerabilities. One key issue found was an SMB/NetBIOS null session on port 445, which could allow unauthorized access to shared resources. This showed how small misconfigurations can turn into serious security risks, especially on critical systems like a domain controller.

What stood out to me was how much information could be gathered just from scanning without direct access. It made it clear that attackers can learn a lot through reconnaissance alone, but also how useful these tools are for defenders to catch and fix issues early.

This lab helped me understand how vulnerability scanning works in a real environment and why it's important for identifying and reducing security risks before they can be exploited.